

Customer Churn & Lifetime Value (CLTV) Prediction

Customer Behavior Analysis using RFM, BG/NBD, Gamma-Gamma & K-Means

Presented by:

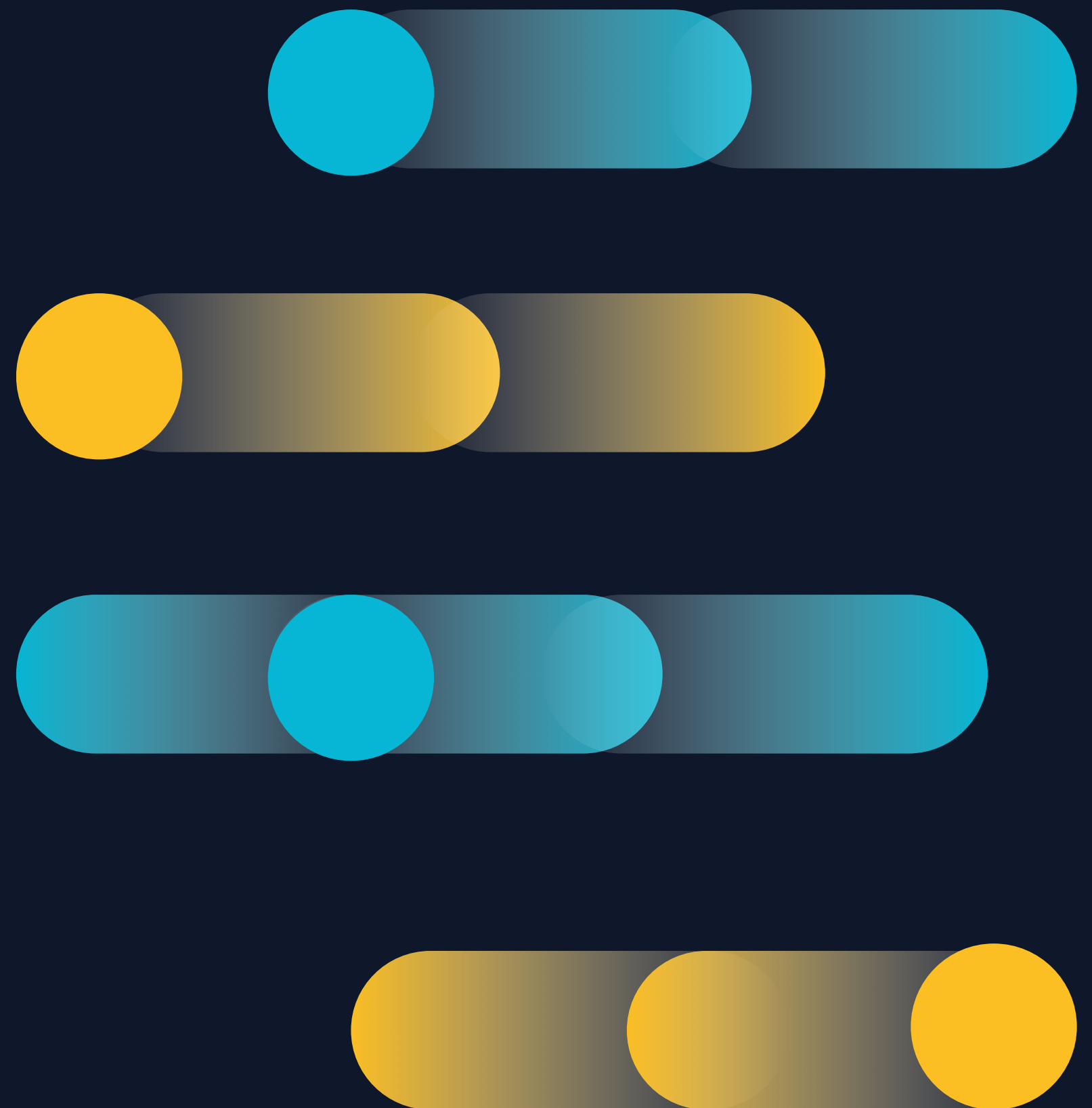
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- **Customer Churn:** When a customer stops purchasing from a company.
- **CLTV:** The estimated value a customer can generate during their relationship with a company.



Problem Statement

E-commerce companies need to understand customer behavior to identify customers who may stop purchasing and customers who generate high future value.

Objectives

- Analyze customer transaction behavior
- Create RFM and Tenure features
- Predict customer churn
- Estimate Customer Lifetime Value
- Segment customers based on behavior



Dataset

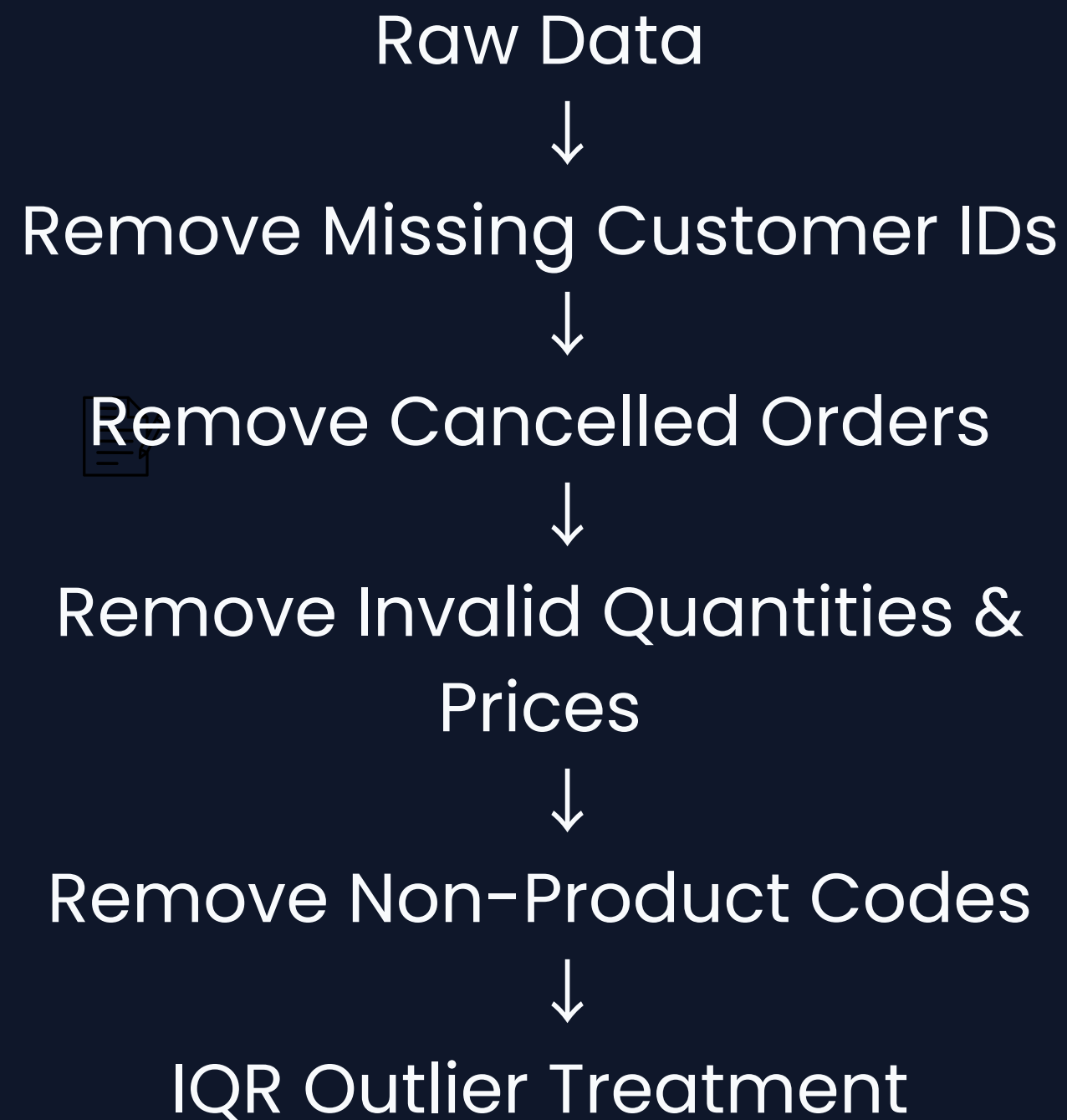


UCI Online Retail II Dataset

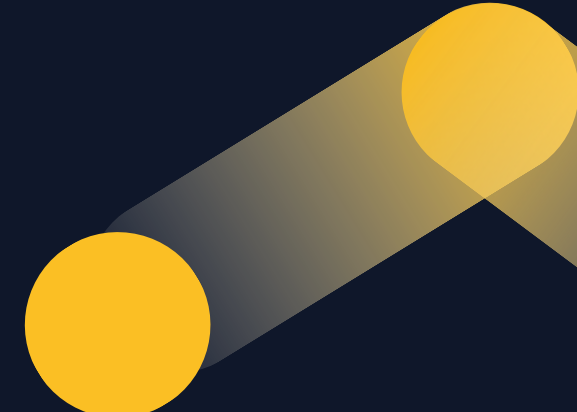
It contains e-commerce transaction information such as:

- Invoice
- StockCode
- Quantity
- Price
- InvoiceDate
- Customer ID
- Country

Data preprocessing



RFM & Tenure Feature Engineering



R — Recency

Days since last purchase

Lower value → more recently active

F — Frequency

Number of unique completed orders

Higher value → more frequent purchases

M — Monetary

Total customer spending

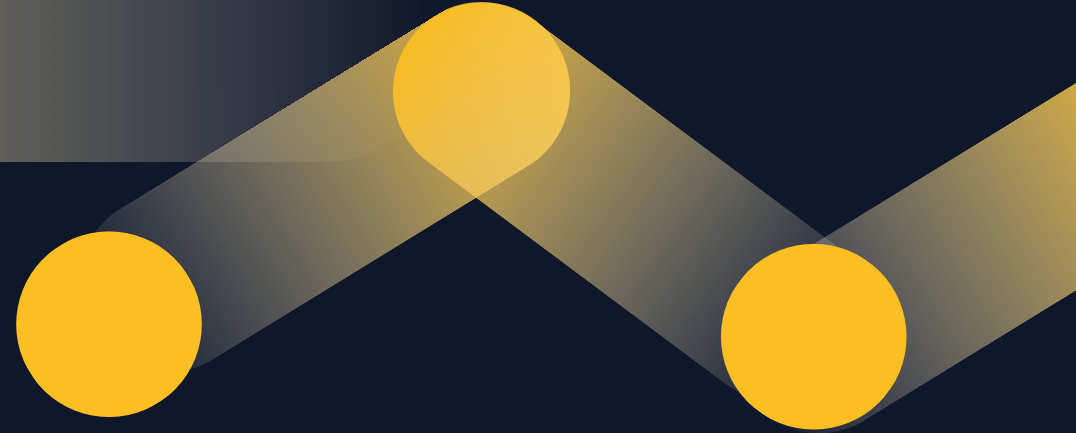
Higher value → higher contribution

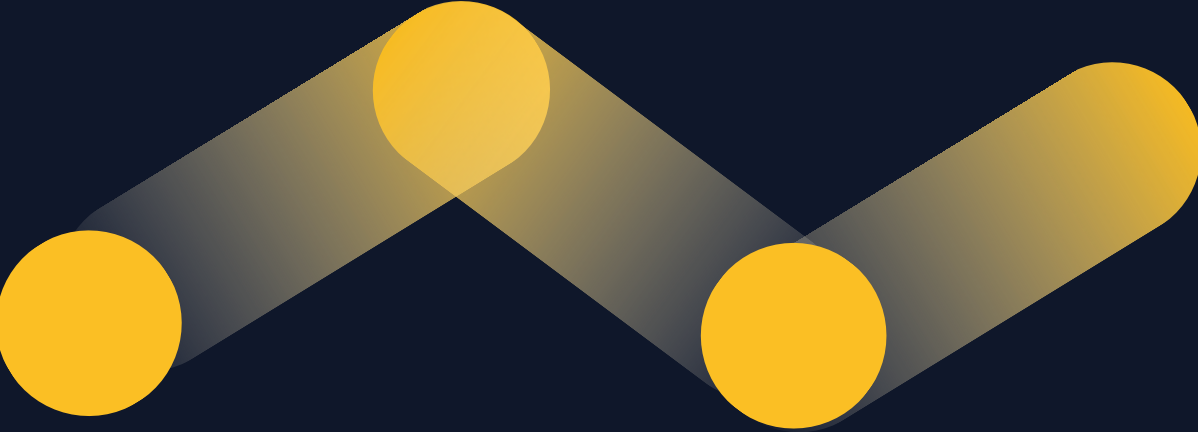
T — Tenure

Days since first purchase

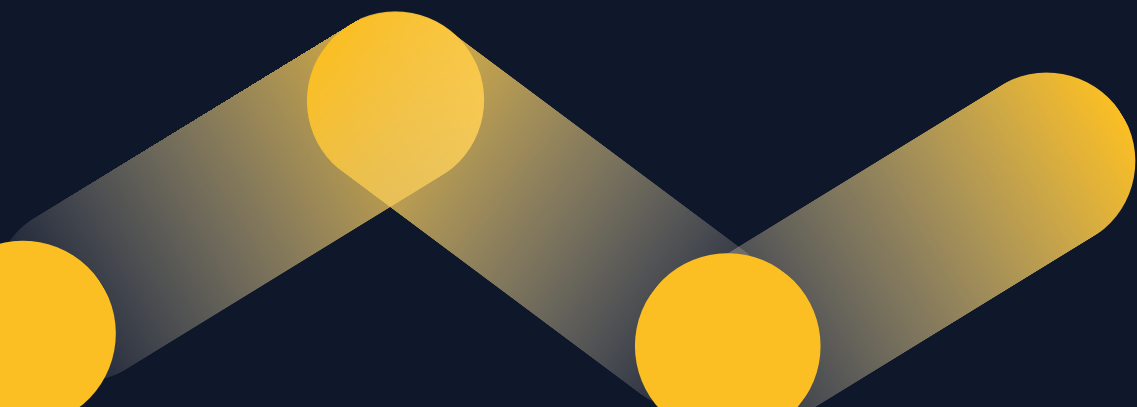
Shows customer relationship duration

4,285 customer profiles generated





Methodology



Churn Prediction using BG/NBD

BG/NBD (Beta Geometric / Negative Binomial Distribution) is a model used to estimate how many times a customer may purchase in the future.

What does it do?

It predicts:

- Future purchases
- Probability that a customer is still active
- Churn risk

- Churn Risk → The likelihood that the customer has stopped purchasing.
- Churn Risk = $1 - \text{Probability Alive}$
- We considered a customer churned when Probability Alive < 0.20 .



CLTV PREDICTION & CUSTOMER SEGEMENTATION

Estimating customer value and grouping customers by behavior

CLTV Prediction

BG/NBD (Predicts future purchases)



Gamma-Gamma Model
(Estimates future spending)



CLTV (Predicted customer value)

CLTV = How much value a customer is expected to generate in the future.

Customer Segmentation

K- Means
(Groups similar customers)



4 Customer Segments

- Champions
- Loyal Customers
- At-Risk
- Hibernating/Lost

RESULTS & KEY FINDINGS

What our analysis tells us about customers

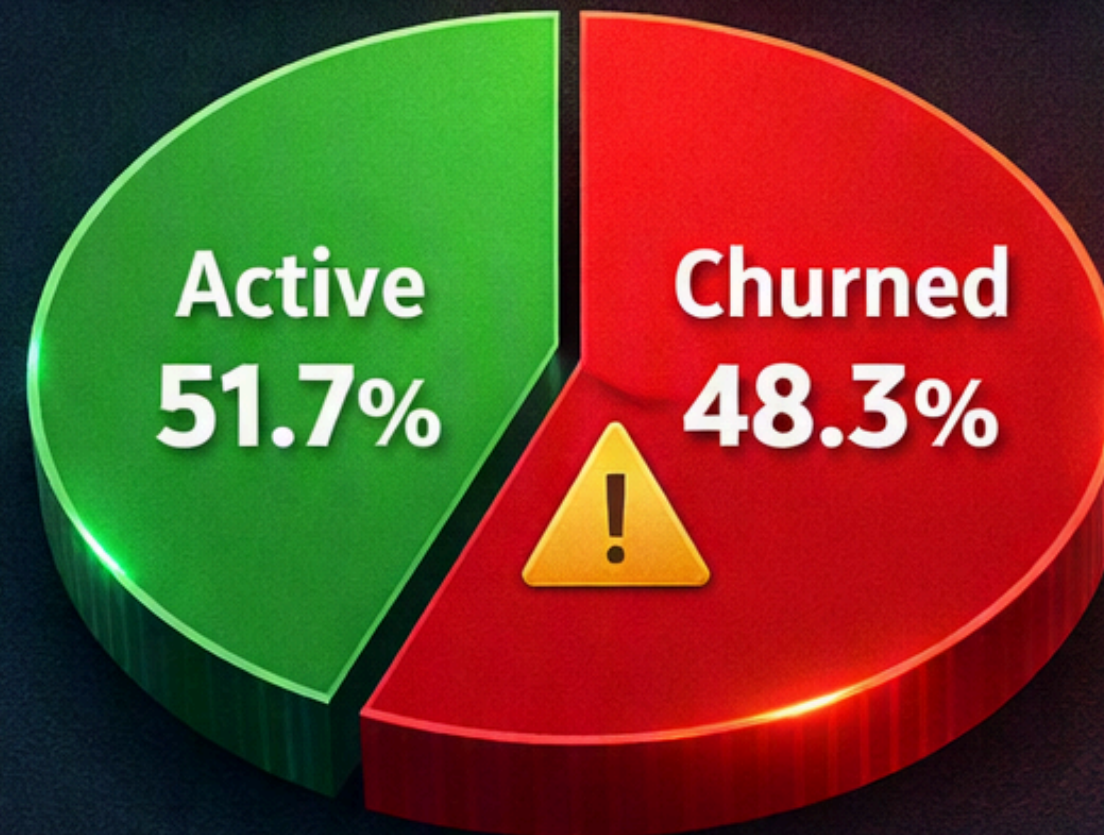
4,285
CUSTOMERS ANALYZED

51.7%
ACTIVE

48.3%
CHURNED

CUSTOMER SEGMENTS

- Champions
- Loyal
- At-Risk
- Hibernating



- ✓ More than half are active
- ⚠ Nearly half show churn risk
- ➡ Customers can be grouped for *targeted strategies*

Technology & Project Outputs

Tools used and outputs generated

Technology Stack

 Python
Main programming language

 Pandas
Data cleaning & analysis

 NumPy
Numerical calculations

 Matplotlib / Seaborn
Data visualization

 Scikit-Learn
K-Means clustering

 Lifetimes
BG/NBD & Gamma-Gamma models

 OpenPyXL
Excel dataset handling

Project Outputs

 RFM Customer Profiles
4,285 customer profiles



 Churn Prediction
Active / Churned classification



 CLTV Prediction
6-month projected customer value



 Customer Segmentation
4 behavioral segments



 Final Dataset
Customer-level predictions & segments

Thank You

